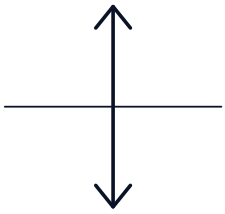
DO NOW · AO1 REVISION

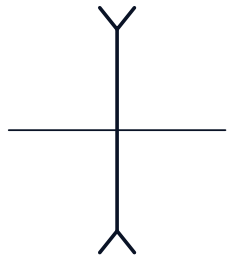
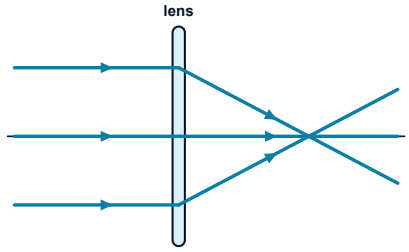
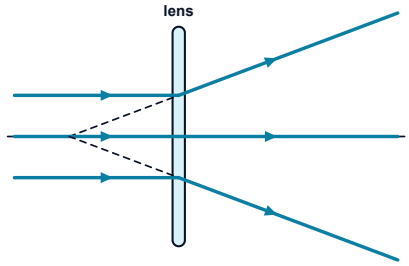
Lenses and visible light

Waves
Paper 2 · Separate Physics only
25 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 6 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**Name the process that happens to light in a lens.**	Refraction	☐
2 ★	**Describe what a convex lens does to parallel rays of light.**	It brings them together at the principal focus.	☐
3	**What is the focal length of a lens?**	The distance from the lens to the principal focus.	☐
4	**Describe what a concave lens does to parallel rays of light.**	It spreads them out, as if from the principal focus.	☐
5 ★	**What is a real image?**	An image formed where rays of light actually meet.	☐
6 ★	**What is a virtual image?**	An image that rays of light only appear to come from.	☐
7	**Name the type of image that can be shown on a screen.**	A real image	☐
8	**Name the type of lens that produces only virtual images.**	A concave lens	☐
9	**Name the type of lens that this symbol represents.** 	A convex lens	☐

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	QUESTION	ANSWER	✓
10	**Name the type of lens that this symbol represents.** 	A concave lens	☐
11 ★	**Name the type of lens shown in this ray diagram.** 	A convex lens	☐
12	**Name the type of lens shown in this ray diagram.** 	A concave lens	☐
13	**State how different colours of light differ.**	They have different wavelengths.	☐
14 ★	**What is specular reflection?**	Reflection from a smooth surface in a single direction.	☐
15	**What is diffuse reflection?**	Reflection from a rough surface, scattering light in many directions.	☐
16	**Explain how colour filters work.**	They transmit some wavelengths and absorb the rest.	☐
17	**Explain why blue objects appear blue.**	They reflect blue light and absorb the other colours.	☐
18	**Explain why white objects appear white.**	They reflect all colours of light equally.	☐

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	QUESTION	ANSWER	✓
19	**Explain why black objects appear black.**	They absorb all colours of light.	☐
20	**In words, state the equation for magnification.**	magnification = image height ÷ object height	☐
21	**State the unit of magnification.**	It has no unit (it is a ratio).	☐
22	**State the types of image that a convex lens can produce.**	Real or virtual	☐
23	**State what determines the colour of an opaque object.**	The wavelengths of light it reflects most strongly.	☐
24	**State what an opaque object does to light.**	It absorbs or reflects it; none is transmitted.	☐
25 ★	**A blue object is viewed in red light. State and explain the colour it appears.**	Black. It absorbs the red light and has no blue light to reflect.	☐

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